

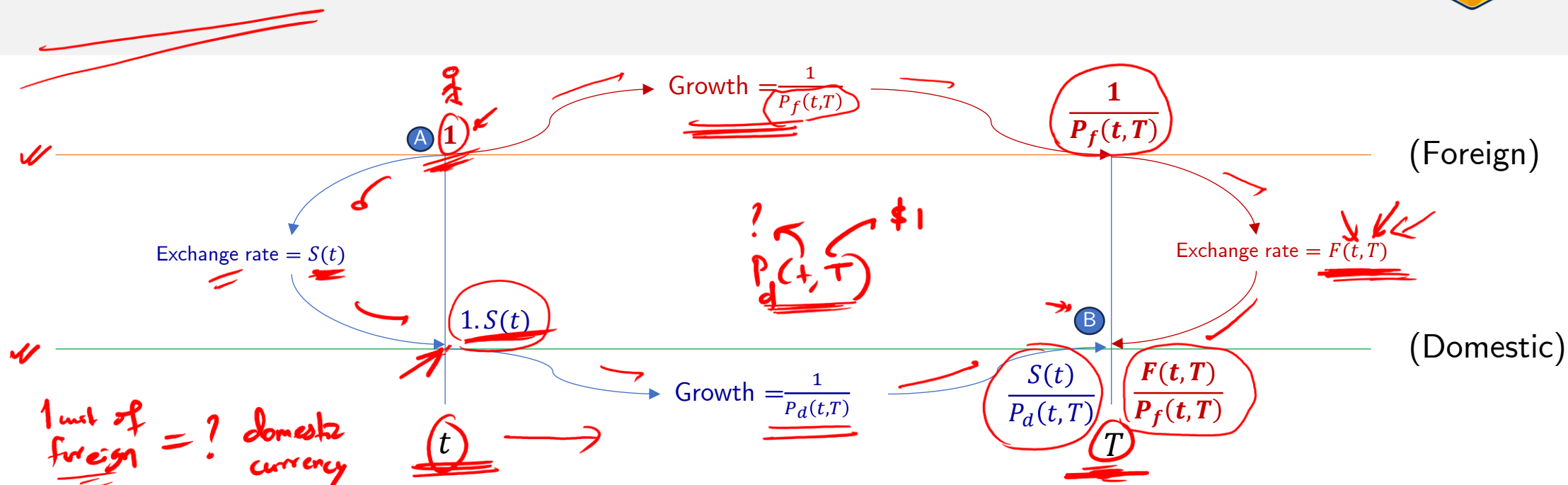
FX Derivatives

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FX Forward rate – The tale of two rivers



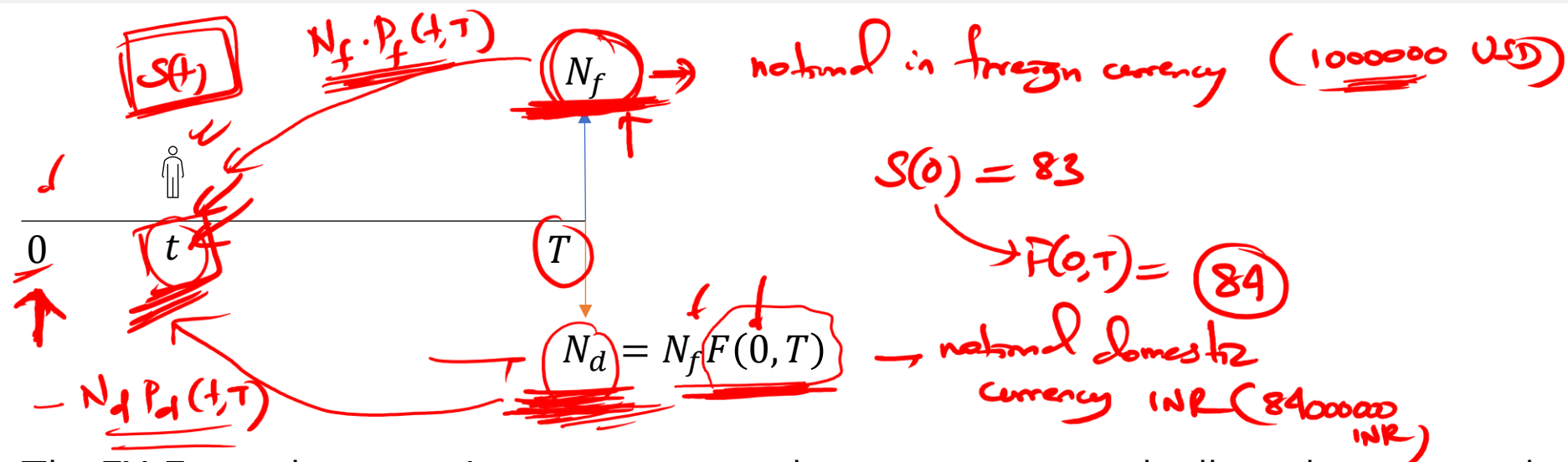
- ▶ You have 1 unit of foreign currency at time t (point A). How much units in domestic currency will you have at T (point B) ?
- ▶ You can invest in Foreign bond maturing at T . If the price of the bond at t is $P_f(t, T)$, the amount of foreign currency at T will be $\frac{1}{P_f(t, T)}$ which can be exchanged to domestic currency using the forward exchange rate $F(t, T)$ which will lead to an amount $\frac{F(t, T)}{P_f(t, T)}$
- ▶ Or you can exchange to domestic currency using the spot rate $S(t)$ which will give you $1 \cdot S(t)$ in domestic currency. Then invest the amount in domestic bond that's priced at $P_d(t, T)$, which will lead to an amount $\frac{S(t)}{P_d(t, T)}$ at maturity T .
- ▶ To prevent arbitrage, we must have

$$\frac{F(t, T)}{P_f(t, T)} = \frac{S(t)}{P_d(t, T)} \Rightarrow F(t, T) = S(t) \frac{P_f(t, T)}{P_d(t, T)} = S(t) e^{(R_d - R_f)(T-t)}$$

$$P_d(t, T) = e^{-R_d(T-t)}$$

$$P_f(t, T) = e^{-R_f(T-t)}$$

FX Forward



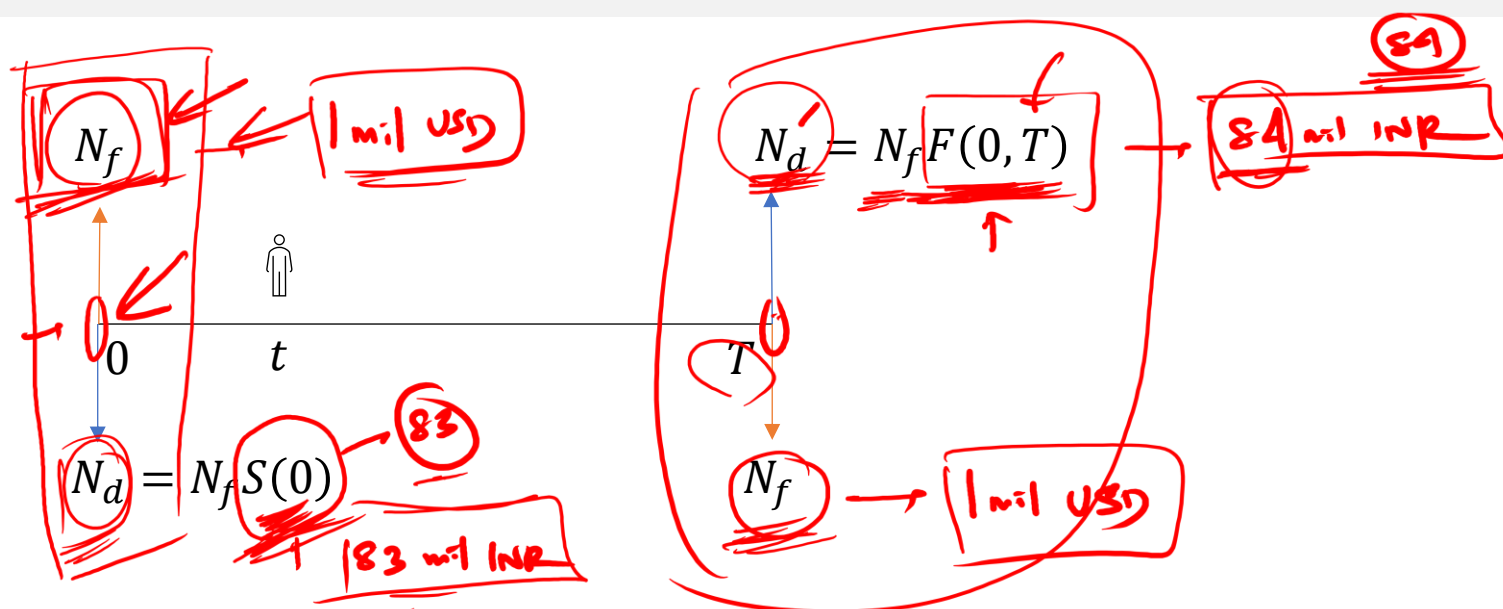
- ▶ The FX Forward contract is an agreement to buy one currency and sell another currency based on a pre-determined exchange rate. This pre-determined rate is nothing but the FX forward rate.

- ▶ The FX forward rate at $t = 0$ is $F(0, T) = S(0) \frac{P_f(0, T)}{P_d(0, T)}$, so $N_f F(0, T) = N_d$

- ▶ The value of the Forward at any time t is $V(t) = (N_f F(t, T) - N_d) P_d(t, T)$ or $V(t) = N_f P_f(t, T) S(t) - N_d P_d(t, T)$

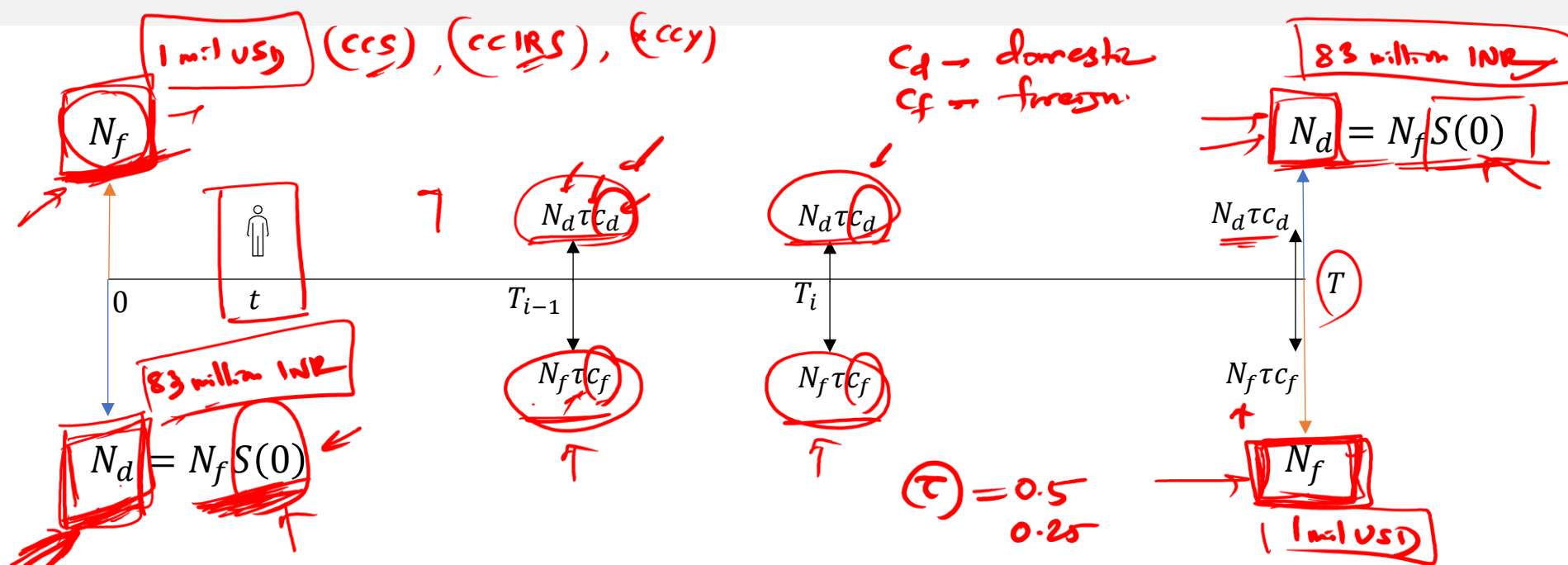
$$= \left(N_f \cdot S(t) \cdot \frac{P_f(t, T)}{P_d(t, T)} - N_d \right) P_d(t, T) =$$

FX Swap



- ▶ The FX Swap involves a **spot transaction** and a **forward transaction**. The spot transaction involves an exchange of currencies at the spot rate. The forward transaction involves reversing the currency exchange at a forward rate.
- ▶ From a valuation perspective, the FX swap is identical to FX forward with the difference of the spot transaction which has a net value of 0.

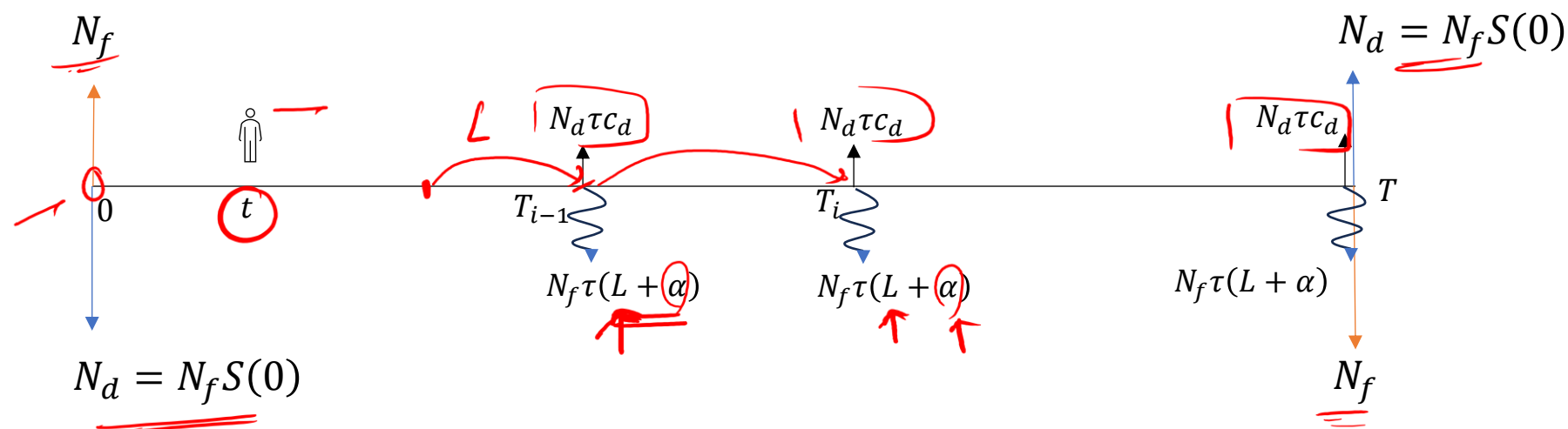
Cross Currency Swap (xCCy) – Fixed/Fixed



► A cross-currency swap (xCCy) is similar to FX swap with 2 key differences

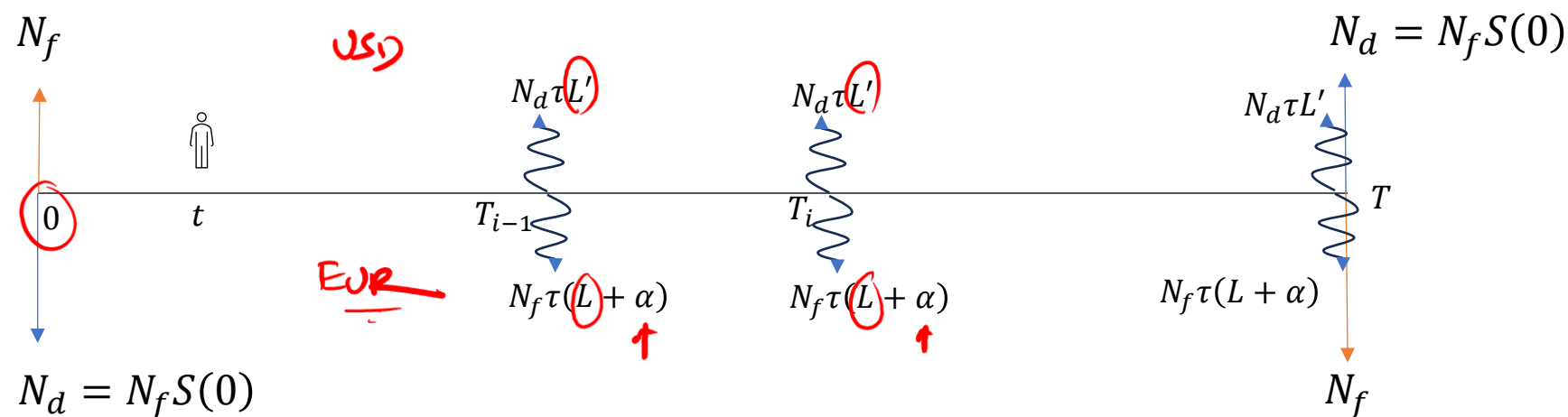
- The far leg is an exact reversal of the near leg i.e. it's based on the initial spot rate and not the Forward rate
 - The contract involves periodic payments of coupons i.e., fixed/fixed, fixed/floating or floating/floating
- The valuation can be done simply using the formulae for Fixed bonds in both currencies. The value of a fixed/fixed xCCy is quoted in terms of PV.

Cross Currency Swap (xCCy) – Fixed/Floating



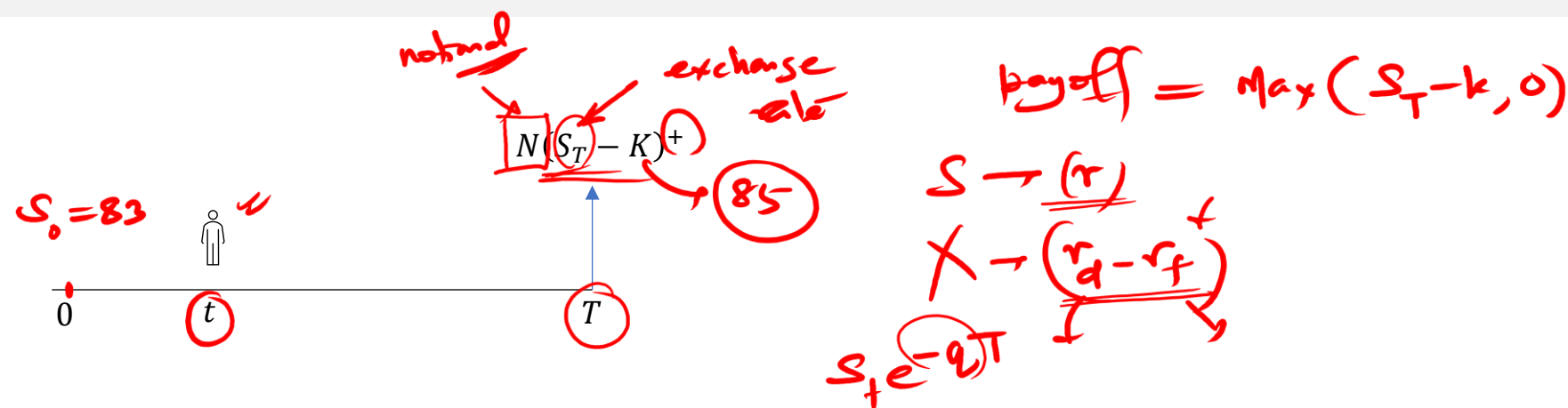
- For a fixed/floating xCCy, we add a spread (α) to the floating leg such that the value at inception is 0.
- This type of contract is conveniently quoted in terms of spread.

Cross Currency Swap (xCCy) – Floating/Floating



- For a floating/floating xCCy, we add a spread (α) to the non-USD leg as convention such that the value at inception is 0.
- This type of contract is conveniently quoted in terms of spread.

FX Option



- ▶ The FX Option works like the equity option except that the underlier is the exchange rate. The pay off of a call option expiring at T is given as Notional $\times (S_T - K)^+$
- ▶ Since the exchange rate grows with a drift $= (r_d - r_f)$, the Black Scholes Formula can be modified slightly to accommodate this drift. The Call Option pricing formula is given as

$$V(t) = S_t e^{-r_f(T-t)} N(d_1) - K e^{-r_d(T-t)} N(d_2) \quad d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r_d - r_f + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \quad d_2 = d_1 - \sigma\sqrt{T-t}$$